TAYLAN ÖZGÜR ÖZDEMIR

Software Engineer (Automation, AI and Systems)

Ankara, Turkey • +90 541 262 3184 • taylan147701@gmail.com • linkedin.com/in/taylan1477 • github.com/taylan1477

EDUCATION

İnönü University *Sep 2021 – Jun 2025*
Bachelor of Science in Computer Engineering, GPA: 3.09 / 4.00 Malatya, Turkey
Coursework: Computer Vision, Data Structures and Algorithms, Distributed Systems, Operating Systems, Machine Learning

WORK EXPERIENCE

SoloRota *Jul 2025 – Jan 2026*
Software & Systems Developer (Full-time) Ankara, Turkey

- Built data ingestion pipelines and REST API integrations for 7 mobile apps published on app stores.
- Profiled system memory and traced thread locks, reducing garbage collection freezes by 25%.
- Automated daily build creation and deployment steps using custom Python and PowerShell scripts.
- Wrote modular data validation schemas and state listeners to keep multi-platform builds in sync.

PROJECTS

Job-Radar: Job Scraping and Candidate Match Engine *github.com/taylan1477/Job-Radar*
Python, Playwright, Pydantic, Scikit-learn, Google Sheets API

- Built a web scraper with Playwright to extract job postings from Single Page Applications while handling anti-bot rate limits.
- Implemented an NLP text-matching algorithm using TF-IDF vectorization and cosine similarity to rank job descriptions against resumes.
- Automated data pipelines to clean, validate with Pydantic schemas, and export job data to Google Sheets and Excel.

Kidney CT Segmentation: Classical CV vs Deep Learning *github.com/taylan1477*
Python, PyTorch, OpenCV, ScaleNorm-UNet, BRISQUE Quality Filter, AMOS Dataset

- Benchmarked classical image processing (HOG, SIFT, CoHOG, LBP) against ScaleNorm-UNet on abdominal CT scans.
- Evaluated input slice quality with BRISQUE scores to automatically filter out noisy and low-contrast scans.
- Trained the deep learning segmentation model with Dice Loss and IoU metrics to identify precise kidney boundaries.

Voice OS (Jarvis): Local Offline AI Desktop Assistant *github.com/taylan1477*
Python, Ollama (Qwen/LLaVA), Faster-Whisper, Win32 IPC, Graphify

- Built an offline voice assistant running local quantized models (Qwen 2.5 7B and LLaVA 7B) on GPU with no cloud calls.
- Wrote a fast speech loop combining Faster-Whisper (INT8) with an interruptible TTS engine that stops speaking when the user cuts in.
- Automated Windows desktop controls via Win32 IPC calls and mapped module dependencies with an AST graph.

TECHNICAL SKILLS

AI and Machine Learning: PyTorch, OpenCV, Computer Vision, ScaleNorm-UNet, Scikit-learn, Ollama (Local LLMs), Faster-Whisper
Scraping and Automation: Playwright, Selenium, Beautiful Soup, Dynamic DOM Extraction, Anti-Bot Bypass, Win32 IPC
Languages and Data: Python, C#, Java, SQL, TypeScript, Pydantic, FastAPI, REST APIs
Systems and Tools: Docker, Linux (Ubuntu/WSL2), Git, GitHub, Bash, PowerShell, Model Context Protocol (MCP)